So, the acute angle between the line and the plane is given by $\theta = \sin^{-1}\left(\frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}|\,|\vec{n}|}\right)$ ... (1)

In Cartesian form if $\frac{x-x_1}{a_1} = \frac{y-y_1}{b_1} = \frac{z-z_1}{c_1}$ and $ax+by+cz=p$ are the equations of the line and the plane, then $\vec{b} = a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $\vec{n} = a\hat{i} + b\hat{j} + c\hat{k}$. Therefore, using (1), the acute angle θ between the line and plane is given by

$$\theta = \sin^{-1}\left(\frac{|aa_1 + bb_1 + cc_1|}{\sqrt{a^2+b^2+c^2}\sqrt{a_1^{\,2}+b_1^{\,2}+c_1^{\,2}}}\right)$$

Remark

(i) If the line is perpendicular to the plane, then the line is parallel to the normal to the plane.

So, $\vec{b}$ is perpendicular to $\vec{n}$. Then we have $\vec{b} = \lambda\vec{n}$ where $\lambda \in \mathbb{R}$, which gives $\frac{a_1}{a} = \frac{b_1}{b} = \frac{c_1}{c}$.

(ii) If the line is parallel to the plane, then the line is perpendicular to the normal to the plane.

Therefore, $\vec{b}\cdot\vec{n} = 0 \Rightarrow aa_1 + bb_1 + cc_1 = 0$

Example 6.48

Find the angle between the straight line $\vec{r} = \left(2\hat{i}+3\hat{j}+\hat{k}\right) + t\left(\hat{i}-\hat{j}+\hat{k}\right)$ and the plane $2x - y + z = 5$.

Solution

The angle between a line $\vec{r} = \vec{a} + t\vec{b}$ and a plane $\vec{r}\cdot\vec{n} = p$ with normal $\vec{n}$ is $\theta = \sin^{-1}\left(\frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|}\right)$.

Here, $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{n} = 2\hat{i} - \hat{j} + \hat{k}$.

So,we get $\theta = \sin^{-1}\left(\frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}|\,|\vec{n}|}\right) = \sin^{-1}\left(\frac{\left|\left(\hat{i}-\hat{j}+\hat{k}\right)\cdot\left(2\hat{i}-\hat{j}+\hat{k}\right)\right|}{\left|\hat{i}-\hat{j}+\hat{k}\right|\left|2\hat{i}-\hat{j}+\hat{k}\right|}\right) = \sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$ ■

6.8.12 Distance of a point from a plane

(a) Equation of a plane in vector form

Theorem 6.20

The perpendicular distance from a point with position vector $\vec{u}$ to the plane $\vec{r}\cdot\vec{n} = p$ is given by

$$\delta = \frac{|\vec{u}\cdot\vec{n} - p|}{|\vec{n}|}.$$

Proof

Let A be the point whose position vector is $\vec{u}$.

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